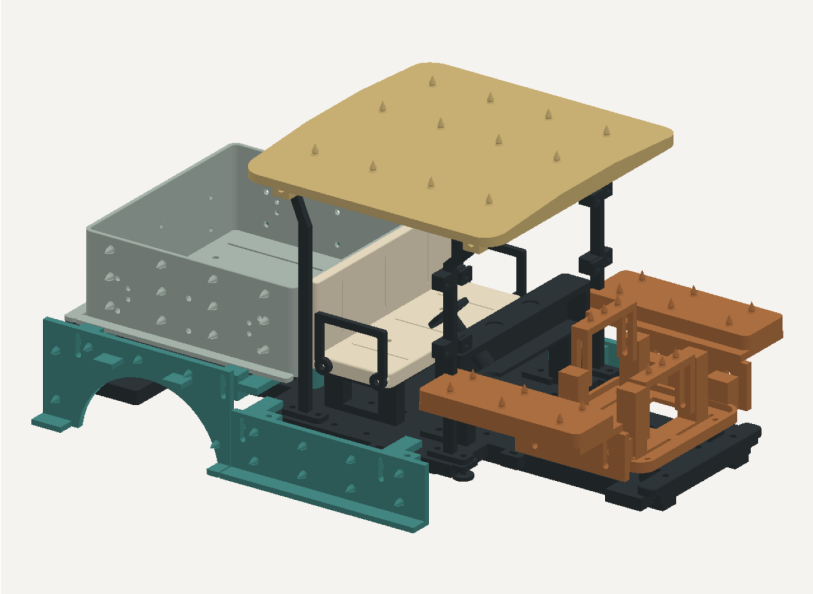
MASTER BUILD GUIDE

IT DEPT / Pumpkin Cart

From printed parts and bare chassis to a complete dry build ready for pumpkin.



Current printed assembly. RC chassis, screws, wiring and clear windshield omitted for visibility.

THIS IS THE CURRENT ASSEMBLY SEQUENCE
Use the successfully printed flat floor, R2 spiked bed/roof, HF2 raised hood carrier and SR1 side runners. Keep your proven spacer-only front chassis mount. If a stage is already assembled, check it and continue; do not dismantle a sound joint just to restart.

Section	Where to start
Prepare all inserts, with installation-face maps	Pages 2-11
Identify parts and sort screws	Pages 12-16
Assemble frame; fit chassis; seat, dashboard and roof	Pages 17-25
Mount module, clamps, bed, hood, fenders and runners	Pages 26-31
Lights, wiring, cargo and final dry-build checks	Pages 32-36
Floor-hole reference	Page 37

Read before using the iron: install the receiving-part inserts before their access is covered. Model coordinates are reference dimensions, not a drilling template. Actual truck fit and pumpkin retention still need physical checks. No pumpkin is attached until the final handoff.

Install the inserts as a batch

Heat-set orientation means the face the iron approaches, not how the part sits on the print bed.

Tick	Part / receiving holes	Total inserts	Map
[ ]	01 rear floor / raised top pads	10	3
[ ]	02 front floor / raised top pads	14	4
[ ]	03 splices / 2 per plate x4	8	5
[ ]	16A optional front + 16B rear adapters	4*	5
[ ]	20C rear bumper blocks / 1 per block x2	2	5
[ ]	06 pedestal / top seat bosses	2	6
[ ]	07 bench / 2 per side	4	6
[ ]	10 front posts / 2 inward ears per post	4	7
[ ]	11 roof / outside clevis cheeks	4	8
[ ]	12 dashboard / angled boss rear end	1	9

CORE INSERTS: 53 / 51 WITHOUT FRONT ADAPTER
* If retaining your direct front spacer-only mounting, omit the two front-adapter inserts. Original chassis fasteners remain separate. Do not insert threads into floor slots, post locating sockets, wiring loops or the unused windshield ear holes.

A **Test one insert first**
Print 00E_M3_insert_fit_test with the same PLA settings. Test 4.0 mm first. The current parts assume a 4.6 mm outside diameter, 5.7 mm long M3 insert; pockets are 6.7 mm deep with screw relief. Confirm your actual insert fit before treating the rest.

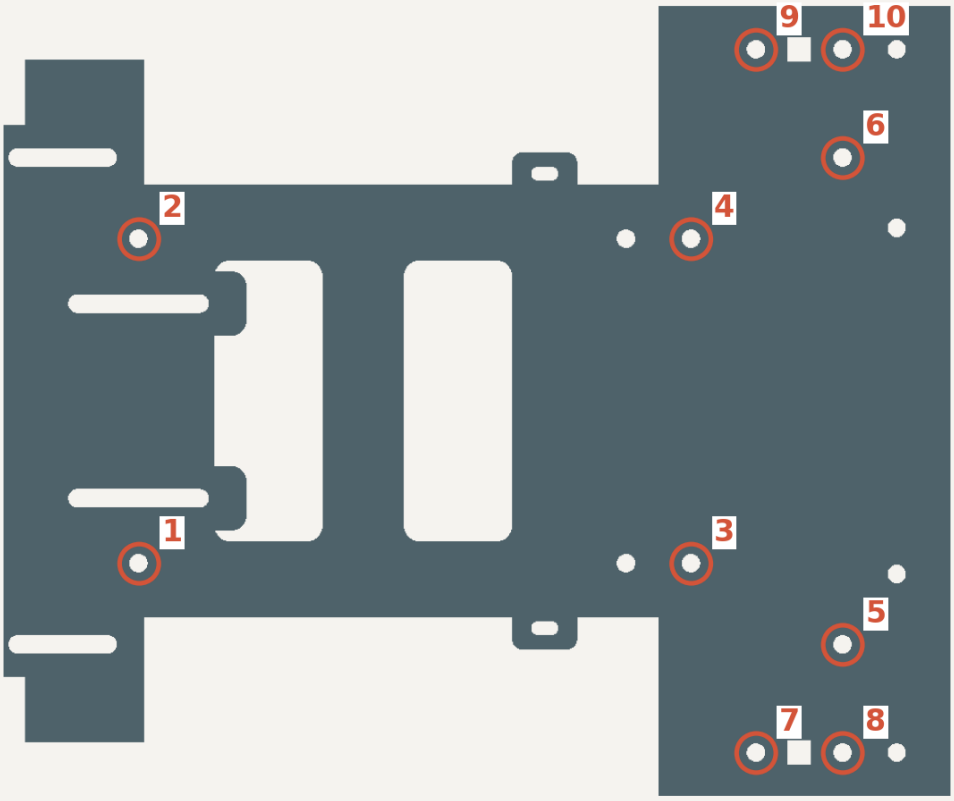
B **Turn each marked entrance face upward**
Support the part so the target bore is vertical and its entrance faces the iron. For side or angled bores, turn the entire part; do not press the iron sideways into a part resting flat.

C **Heat, seat flush, then cool**
Use an insert tip and a controlled PLA-compatible iron setting tested on the coupon. Press slowly along the bore axis, stop flush, and hold square while cooling. Avoid bulging or pushing through. After cooling, hand-thread a screw to verify the thread is clear.

A larger visible circular pad is a reinforcement, not an instruction to insert every hole. Only the red-marked entrances on the following maps receive inserts.

01 / Rear floor: 10 inserts

View from above. Flat mounting face toward the iron; long underside rails face away from it.



Red numbers	Connection	Hole centers in CAD mm
1-4	Rear bed	X=-150 and -48; Y=+/-30
5-6	Rear pair of pedestal feet	X=-20; Y=+/-45
7-10	Rear roof-post feet	X=-36 and -20; Y=+/-65

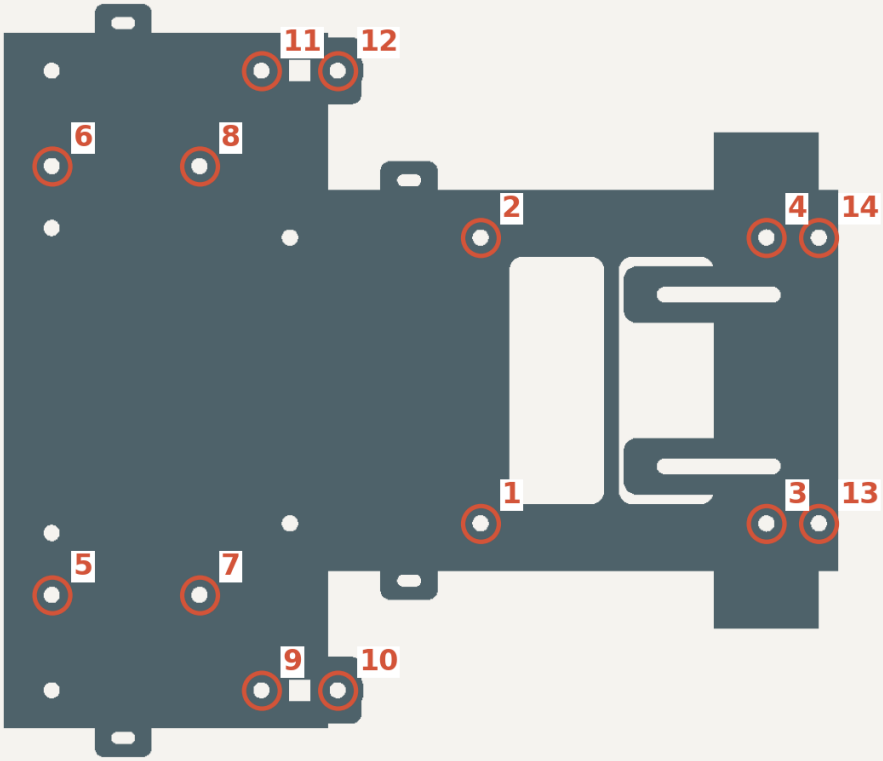
D **Install straight down from the top**
Keep the flat mounting face upward. Seat one insert flush in each of the ten red-marked pockets. The insert axis points down through the floor, toward its underside.

LEAVE ALL OTHER OPENINGS EMPTY
The center splice clearance holes receive screws only. The inner long chassis slots and outer rear bumper slots receive screws only; their threads live in separate parts. The square roof-post sockets and zip-tie eyelets never receive inserts.

Diagram numbers identify holes on this page only. Rear is the outside end; the straight edge facing the other floor half is the center seam.

02 / Front floor: 14 inserts

View from above. Same installation direction as the rear floor: iron toward the underside.



Red numbers	Connection	Hole centers in CAD mm
1-4	Hood carrier spacers	X=100 and 160; Y=+/-30
5-6	Front pair of pedestal feet	X=10; Y=+/-45
7-8	Dashboard feet	X=41; Y=+/-45
9-12	Front roof-post feet	X=54 and 70; Y=+/-65
13-14	Front bumper	X=171; Y=+/-30

E

Install the fourteen marked pockets

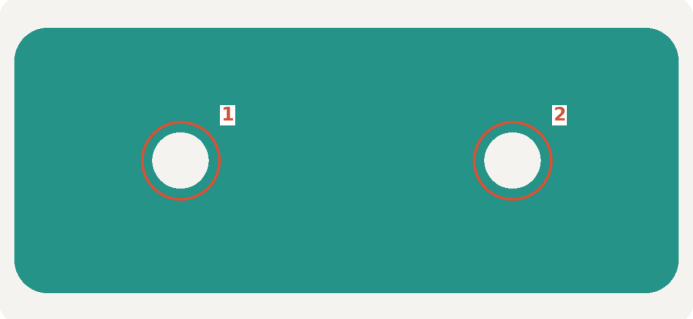
For INSERT installation, keep the flat mounting face upward and rails downward. All fourteen inserts enter the assembled top mounting face and seat flush. No floor insert is installed from underneath.

DO NOT INSERT INTO EVERY ROUND HOLE

The splice screw holes and old unused central chassis holes stay as clearance holes. Long chassis slots stay empty. Only the red-ringed pockets get inserts.

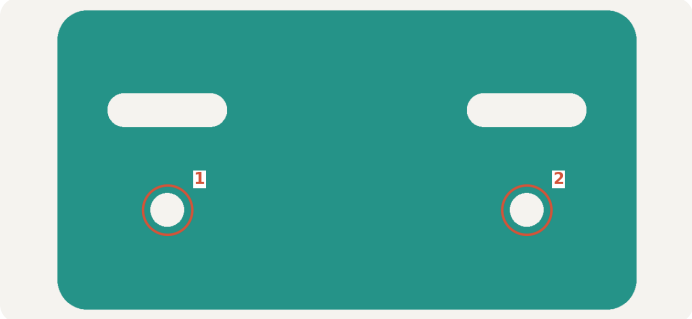
Plates and blocks: 14 inserts

All four views look at the insert installation face. Put that face upward when using the iron.



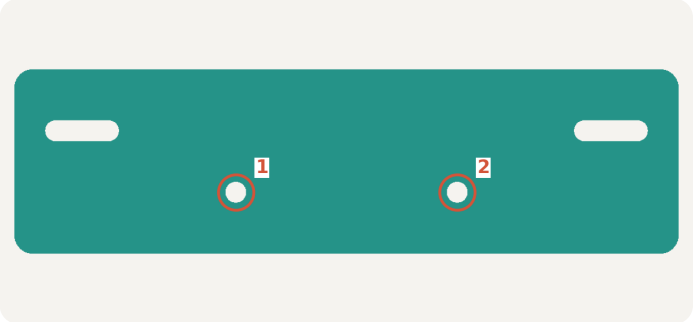
03 / Four splice plates

2 inserts per plate = 8. Enter the top face that will touch the floor.



16A / Front adapter OPTIONAL

2 inserts. Round row only, from the top. Omit this plate if keeping your direct front mount.



16B / Rear adapter

2 inserts. Round row only, from the top. Chassis slot row remains empty.



20C / Two rear slider blocks

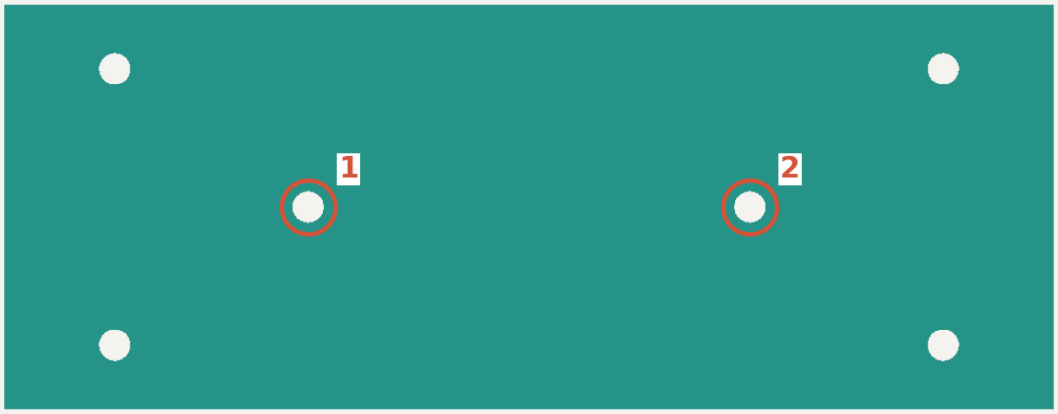
1 insert each = 2. Enter the top face that will touch the underside of the bumper slot.

SPLICES AND BLOCKS MUST BE DONE BEFORE ASSEMBLY
Their insert mouths are covered by the floor once mounted. Insertion points into the part; after cooling, orient the same face upward for assembly. The floor screw passes down into the preinstalled insert.

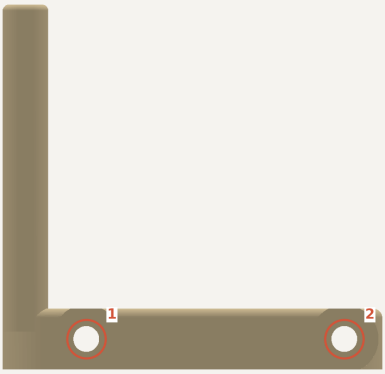
Core modeled total on this page is 14. With no front adapter, install 12. Part 17 mounting shims remain plain clearance spacers and contain no inserts.

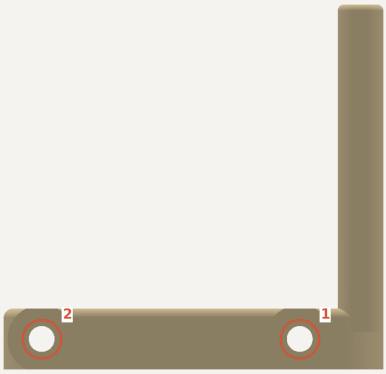
Seat parts: 6 inserts

The pedestal has top-facing pockets. The bench has side-facing pockets; turn it for each side.



06 PEDESTAL: two inserts. Install from the top seat-support plate, into the two red-ringed bosses. These hold the bench. The four floor-fastening holes in the pedestal receive screws only.





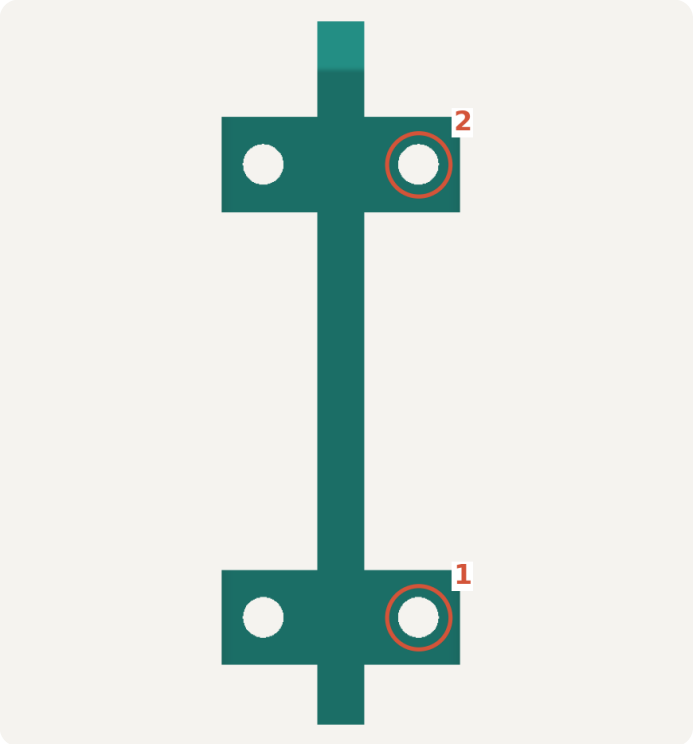
07 BENCH: four inserts, two on each side. Views show the two exterior cushion sides. Turn the bench so one exterior side faces upward and its two holes are vertical. Install two inserts inward toward the hollow cushion, then flip the bench and repeat on the other side.

> **ARMRESTS AND BENCH BASE DO NOT GET INSERTS**
> The armrest lugs stay as screw clearance holes. The two vertical bench compression-boss holes also stay clear: screws pass through them into the pedestal inserts. Do not install bench side inserts from inside the cushion.

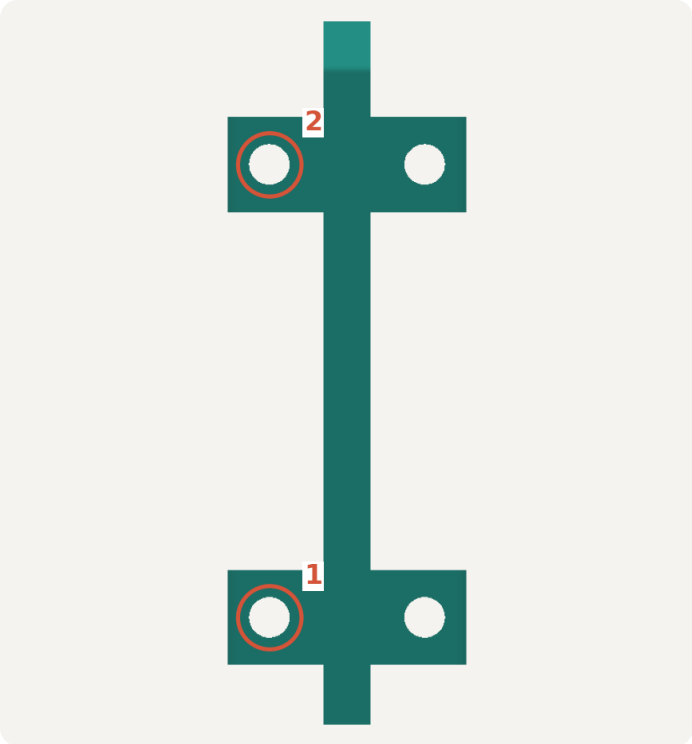
The original armrests can be reused. Side mounting uses M3x8; bench-to-pedestal uses M3x16. Install all six inserts before mounting the bench or armrests.

Front post ears: 4 inserts

Both parts are number 10. Only the two ear holes closest to the center of the cart are used.



LEFT POST / Y=+65
Two inward holes, local Y=-6.5.



RIGHT POST / Y=-65
Two inward holes, local Y=+6.5.

F **Approach from the seat side of each ear**
These views look from the seat toward the hood. Insert entrances are on the rear face of the ears. Turn the loose post until this face points upward; push along the hole axis toward the hood side.

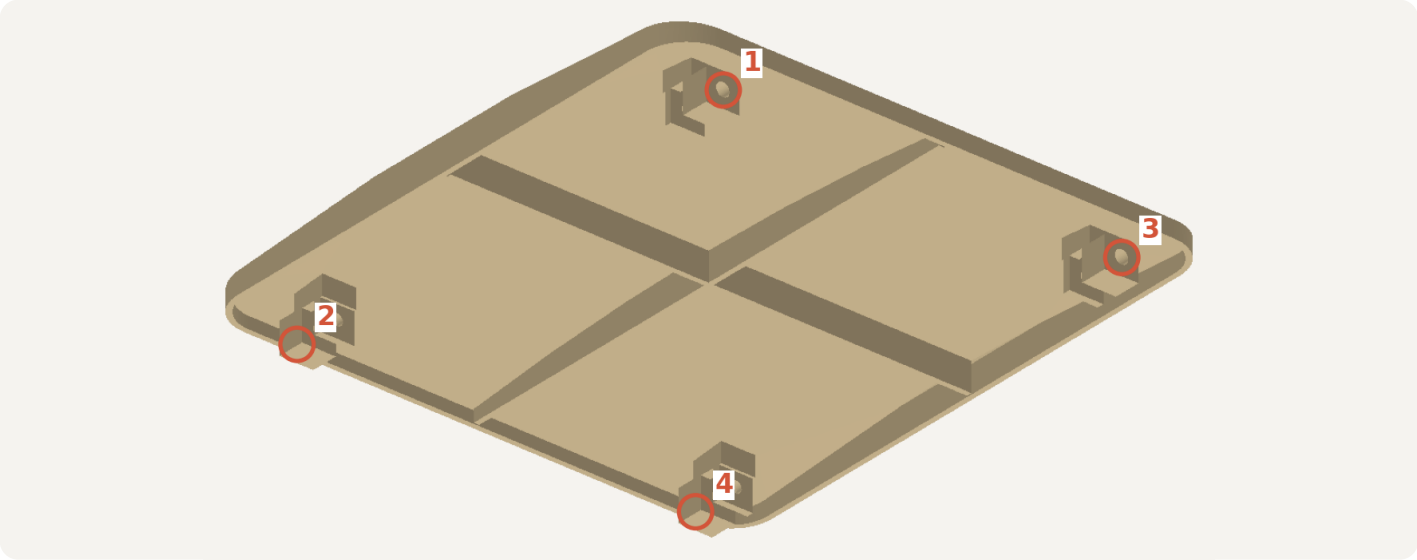
G **Install two per post, leave the outer holes empty**
One insert goes in the lower inward ear and one in the upper inward ear. The two spare outer ear pockets on each post stay empty. Foot holes and the top roof-tab cross-hole are clearance holes.

SCREWS ENTER FROM THE OPPOSITE FACE
Later, windshield M3x10 screws enter from the hood side through the pressure plate and clear sheet. They thread into these inserts. Install inserts before the posts, windshield or roof restrict iron access.

The two posts are identical loose prints. Their installed positions determine which ear holes are inward. Use the red rings above rather than filling all four pockets on each print.

Roof clevis cheeks: 4 inserts

One insert per corner, in the thicker OUTSIDE cheek. The thin inside cheek stays clear.







Side views: two red entrances on each outer roof edge. The underside view shows all four corner positions. Each bore is horizontal across the cart when the roof is assembled.

H **Turn the roof onto its side for installation**

Support the roof so one outside edge faces upward. Enter the thicker outer cheek from outside, pressing toward the cart center. Install both inserts on that edge, then turn the roof and repeat on the opposite edge.

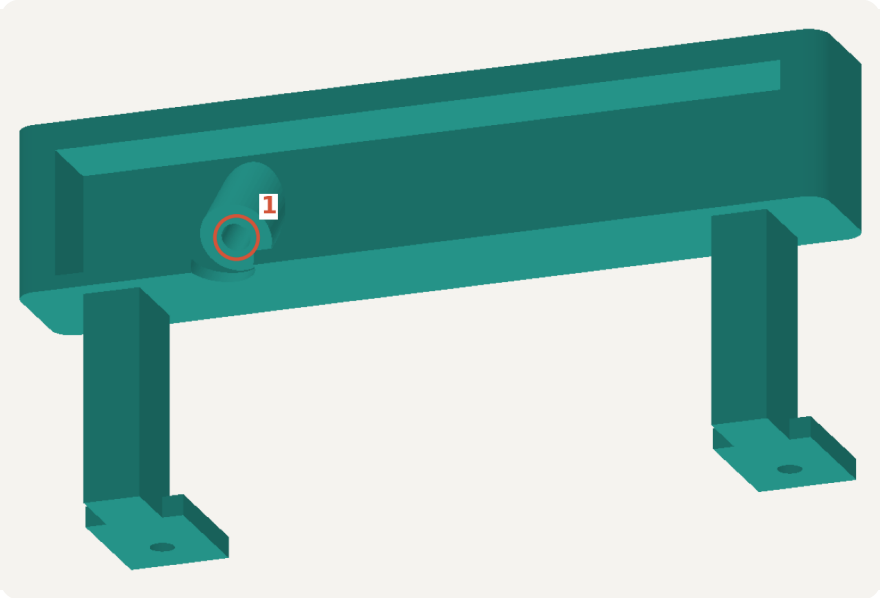
CHECK WHICH CHEEK IS THICKER

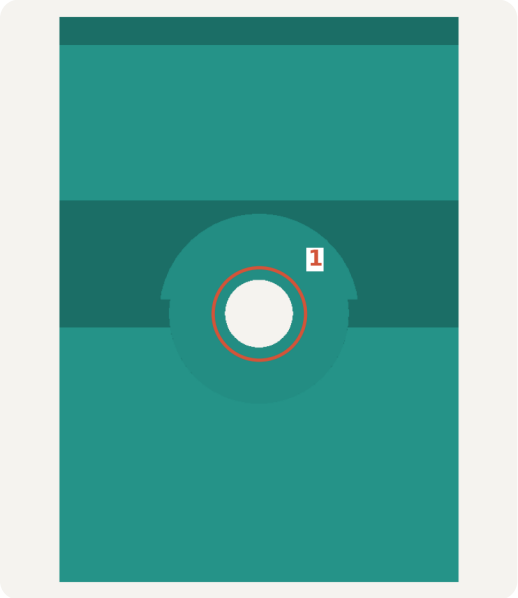
The insert cheek is 8 mm thick; the opposite inner cheek is 2 mm thick. Do not heat an insert into the thin cheek, the post tab or the square locating socket. Keep the clevis gap clean.

During assembly, each M3x16 cross-screw starts at the thin inward cheek, passes through the post tab, and threads into the thick outward cheek. The screw head faces the center of the cart; the insert entrance faces outside.

Dashboard boss: 1 insert

Install into the rear end of the angled steering boss, accessed through the open hood-side cavity.





I **Find the small red-ringed rear entrance**
The rear access relief provides a straight approach for the insert tip. The large socket on the seat-facing side accepts the steering-column stem. It is not an insert pocket. The insert belongs at the opposite, rear end of that angled bore, inside the open dashboard cavity.

J **Aim the bore vertically before pressing**
Tilt the loose dashboard until the rear insert entrance faces upward and the angled bore points straight down. Press from the rear toward the steering-wheel side, along the bore axis; do not press straight down relative to the dashboard feet.

ASSEMBLY DIRECTION
After cooling, the M3x35 screw enters through the steering wheel and column from the seat side, then threads into the rear insert. The two dashboard-foot holes stay clear; those screws use the front-floor inserts.

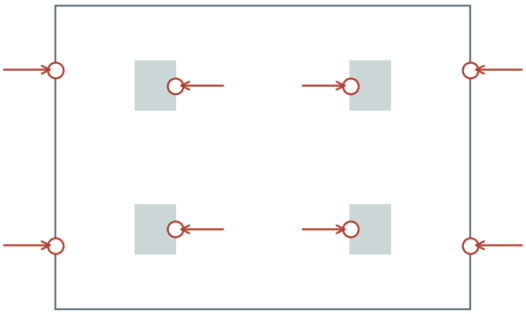
Final batch check	Confirm before moving to assembly
Core counts	24 floor + 8 splice + 4 adapter* + 2 slider + 2 pedestal + 4 bench + 4 front post + 4 roof + 1 dash = 53. *51 with no front adapter.
All entrances	Insert flush, aligned with bore, fully cool; no bulging, cracked PLA, blocked threads or loose movement.
Clearance holes	Slots, sockets, screw pass-through holes and zip-tie eyelets remain free. Start test screws by hand after cooling.

Hood supports: 12 inserts

HF2 parts are separate from the similarly numbered main-cart parts.

HF2 01: TOP VIEW

FRONT +X



4 inner tower mouths: Z32
4 outer boss mouths: Z12

HF2 04: PAD UNDERSIDE



2 inserts per pad x 2 pads
Install into page, toward spikes

Receiving part	Qty	Iron enters from
HF2 01 / outside fender bosses	4	OUTSIDE toward center; two per side. Centers local X +/-22, Z12.
HF2 01 / raised inner towers	4	INSIDE toward outside; two per side. Centers X +/-18, Z32.
HF2 04 / two fender pads	4	UNDERSIDE toward spikes; two per pad at local X +/-22.

ORIENTATION MAP
Carrier bottom is local Z0. The hood faces +X. On each side, the low outer bosses accept fender bolts; the tall inner towers accept spike-bar bolts. Install both sets before fitting the bars. On each pad, flip the flat underside upward for the two inserts.

K

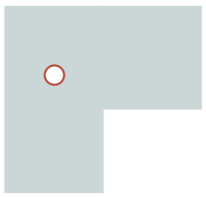
Leave the moving pieces unthreaded
HF2 02/03 riser slots and both HF2 05 spike bars receive screws only. Do not install inserts into their slots. The four base mounting holes remain clear; they use the existing front-floor inserts.

USE THE HF2 CARRIER
HF2 has four tall towers and TWO separate sliding spike bars. The older carrier with low spikes is superseded. Both fender pads and risers from HF1 remain compatible.

Side runners: 12 inserts

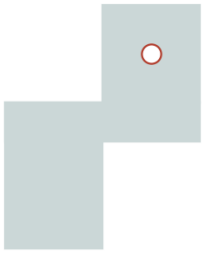
Fit these before closing clamps or mounting the bed hangers.

03 UPPER: OUTSIDE



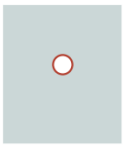
1 per clamp x 4
Insert into page

04 LOWER: TOP



1 per jaw x 4
Insert downward

05 / 06: OUTSIDE END



1 per hanger x 4
Insert into page

Receiving part	Qty	Installation direction
SR1 03 upper cab clamps	4	OUTWARD side face -> cart center. One per clamp.
SR1 04 lower jaws	4	TOP face of offset ear -> downward. One per jaw.
SR1 05/06 bed hangers	4	OUTWARD end boss -> cart center. One per hanger.

L

Turn each target face upward

Put the loose part on a stable support with its insert entrance upward. Approach along the bore, seat flush, then let it cool before starting a screw. The small hole in the lower jaw is at the offset screw ear.

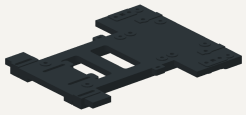
NO INSERTS IN THE PANELS

SR1 cab skirts and rear arches have clearance slots only. Bed spacer holes and optional outward-offset spacers remain clear. All 12 SR1 inserts live in the brackets, not in the pumpkin panels.

Complete current build: 53 core + 12 HF2 + 12 SR1 = 77 inserts with both end adapters. Your spacer-only front mount omits two adapter inserts: **75 total**, assuming the rear adapter is retained. Do not reinstall inserts already fitted.

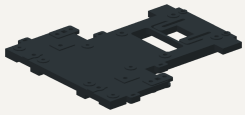
Frame parts checklist

Part numbers match the starts of the STL filenames. Print at 100% scale in millimeters.



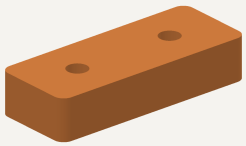
01 / Rear floor

1 copy. Under the cargo bed; larger rear end crosspiece.



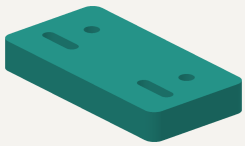
02 / Front floor

1 copy. Under the hood and dashboard.



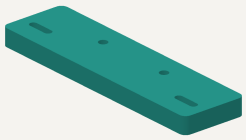
03 / Splice plates

4 copies. All four are the same 8 mm-thick insert part.



16A / Front adapter

Optional. Keep your fitted spacer-only front mount.



16B / Rear adapter

1 copy. Wider; nominal 86 mm chassis-hole spacing.



17 / Mounting shims

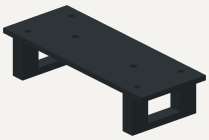
4 copies. 12 mm tall; two per end adapter.

FIT PIECES AND OPTIONAL SHIMS
00 / 00b: socket coupon and 4 mm peg; check the locating fits.
00C / 00D: 2 mm front/rear mounting templates; never use as load-bearing adapters.
17B: optional 2 mm height shim. Add only where the physical fit requires it.

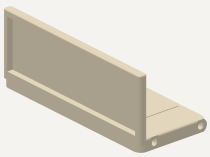
Do not substitute the old v0.4 parts 03, 16 or 17. The new floor uses four splices, separate front/rear adapters and taller shims.

Upper cart parts

Core parts below are reused. The current bed, roof and body supports are listed next.



06 Seat pedestal
Print 1



07 Bench seat
Print 1



08 Armrests
Print 2



09 Rear posts / short bend
Print 2



10 Front posts / long bend
Print 2



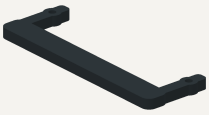
12 Dashboard
Print 1



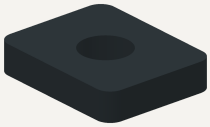
13 Wheel + column
Print 1



15 Hood spacers / 9 mm
Print 4



20B Rear bumper
Print 1

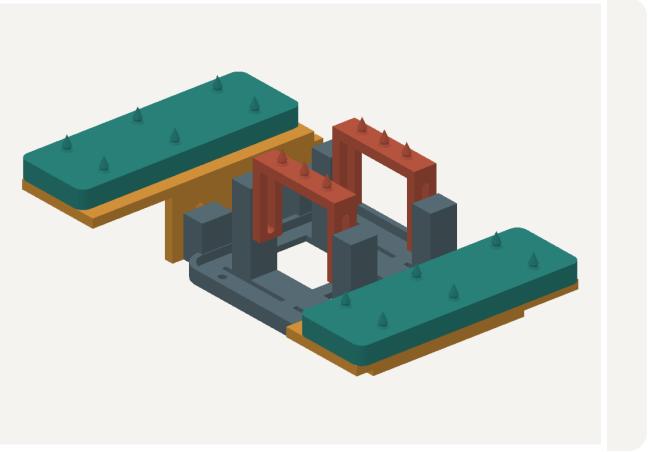


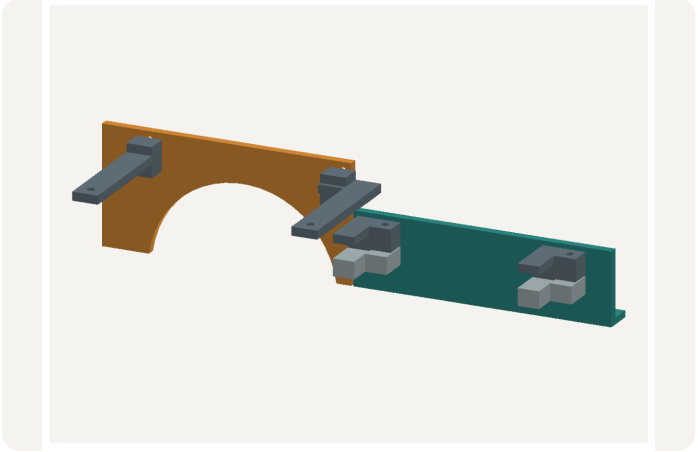
21 Windshield plates
Print 4

SMALL FINISHING PARTS
18: two LED bezels. **19:** two IT DEPT plaques. Also supply a clear windshield sheet, real pumpkin pieces, soft thread/ties, and a removable moisture barrier. Also print one 20A front bumper (or reuse the old one) and two 20C rear insert blocks. Decorative cargo is optional.

Identify the upgrade pieces

Prefixes matter: HF2 01 is not main-cart 01, and SR1 04 is not the bed.





Package	Required pieces
R2 bed / roof	One 04_bed_50mm_sharp_6mm_R2; one 11_roof_sharp_6mm_R2 if using spiked roof. Original roof mounts remain unchanged.
HF2 hood + fenders	01 carrier x1; 02/03 mirrored risers x1 each; 04 pad x2; 05 sliding spike bar x2. Reuse four original 9 mm hood spacers.
SR1 side panels	01 cab skirt left/right x1 each; 02 rear arch left/right x1 each.
SR1 brackets	03 upper clamp left/right x2 each; 04 lower jaw left/right x2 each; 05 rear hanger left/right x1 each; 06 forward hanger left/right x1 each.
SR1 spacers	07 bed spacers, 16 mm x4. Optional 08 (5 mm) or 09 (10 mm) stand-offs: two per panel being offset.

SET ASIDE THE SUPERSEDED PIECES
Do not fit original main-cart 14 hood carrier or HF1 low-spike carrier. For SR1 rear hangers, replace the four 20 mm bed spacers with 16 mm spacers plus the 4 mm hangers. Keep the four 9 mm hood spacers.

Use the successfully printed flat-floor revision. The R2 bed has 50 mm walls; its seat-facing wall has no pumpkin spikes. If a roof or bed is still the earlier version, its mounting interface is reusable, but it will not provide the current spike retention.

Hardware and tools

M3 means 3 mm nominal thread diameter. Listed lengths are measured under the head, not including it.

Connection	Starting bolt	Qty
All floor splices	M3 x16	8
Frame to adapters, where used	M3 x25	4
Bed + 16 mm spacers + 4 mm hangers	M3 x30	4
HF2 base + 9 mm spacers	M3 x16	4
Seat pedestal to floor	M3 x30	4
Bench to pedestal	M3 x16	2
Armrests to bench	M3 x8	4
Dashboard feet	M3 x8	2
Steering column	M3 x35	1
Roof-post feet	M3 x8	8
Roof cross-bolts	M3 x16	4
Windshield pressure plates	M3 x10	4
Front bumper	M3 x10	2
Rear bumper slider blocks	M3 x20	2

CORE ONLY (before HF2/SR1 additions): 53 screws, up to 53 inserts, and one washer per head where it seats flat. M3 lengths: x8 (14), x10 (6), x16 (18), x20 (2), x25 (4), x30 (8), x35 (1). The optional front adapter accounts for two inserts and two x25 screws.

> **CHASSIS HARDWARE STAYS AS FITTED**
> Match the actual truck mounting threads and usable depth. Keep your front spacer-only mounting arrangement. The adapter illustration is a reference option, not a requirement to add the front plate.

Tools: soldering iron with an insert tip, matching drivers, deburring tool, ruler/calipers, flush cutters, and hardware trays. Listed lengths assume typical washers; confirm engagement and tip clearance. Tighten gently by hand after inserts cool.

Additional screws and total kit

Use head washers wherever they sit flat; do not bury screw heads in PLA.

Joint	Fastener	Qty
HF2 carrier -> original floor inserts	M3 x16, already in core count	4
HF2 sliding spike bars -> towers	M3 x10 + washer	4
HF2 fender risers -> carrier sides	M3 x10 + washer	4
HF2 fender pads -> riser shelves	M3 x10 + washer	4
SR1 clamp upper -> lower jaw	M3 x16 + washer	4
SR1 four panels -> brackets, no offset	M3 x10 + washer	8
Bed -> hanger + 16 mm spacer -> floor	M3 x30, already in core count	4

TOTAL PRINTED-KIT SCREWS
At the innermost SR1 panel setting: **77 screws** with both adapters, or **75** with your front adapter omitted. Original truck/direct-mount fasteners are additional and stay as fitted. Full modeled quantities: x8=14, x10=26, x16=22, x20=2, x25=4, x30=8, x35=1. Omit two x25 if the front adapter is omitted.

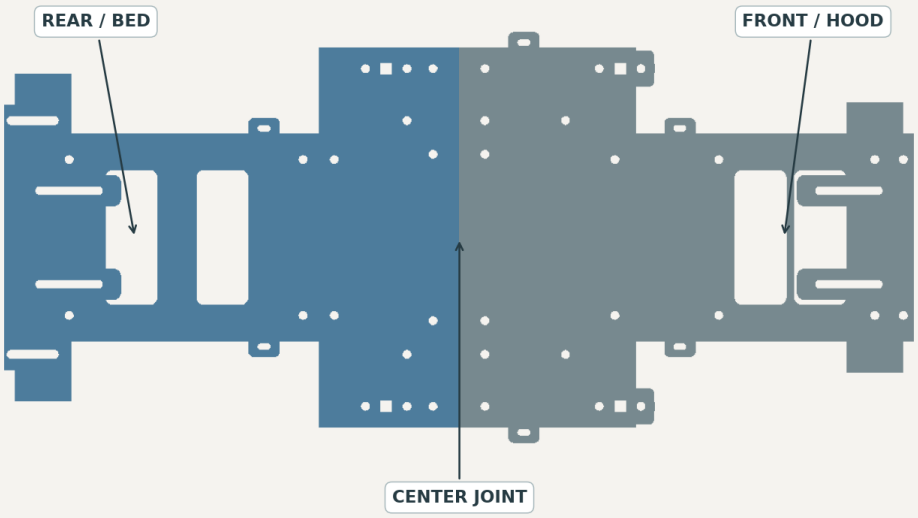
SR1 outward offset	Panel screw / washer stack
0 mm	M3 x10 + about 0.5 mm washer
5 mm	M3 x16 + 1.5 mm TOTAL washer thickness
10 mm	M3 x20 + about 0.5 mm washer

DO NOT FORCE A BOTTOMED SCREW
These nominal stacks leave about 5.5 mm engagement in an assumed 5.7 mm insert. Check actual washers, inserts and screw tips. A screw that gets tight while the part remains loose may have bottomed. Use the correct length instead of tightening harder.

The new parts add 24 inserts and 24 screws to the original kit. No new loose nuts are required. The fit coupon remains the first check for an unfamiliar insert.

Orient and clean the floor

Rear = cargo bed. Front = steering/hood end. The broad center area is the cab.



1. **Place 01 and 02 top-side up**
   For assembly, the flat mounting face and insert entrances face up. The two long reinforcing rails face down. Bring the straight center edges together; do not flip one half upside down.

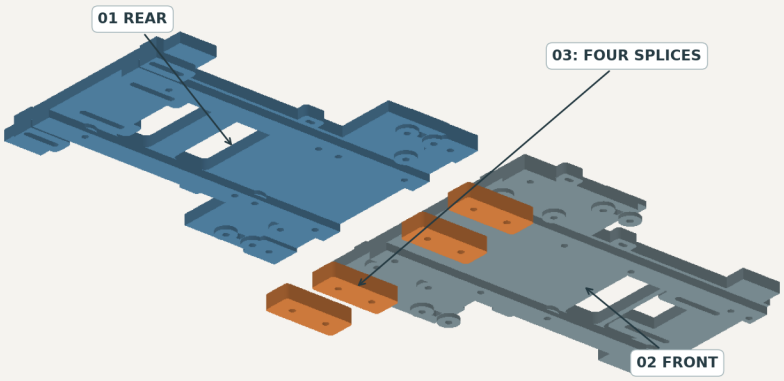
2. **Remove supports and check mating faces**
   Clean the center edges, bolt holes, slots and underside splice contact areas. A straightedge should sit across the joint without a large step or rocking.

3. **Sort the underside connections**
   There are four splice locations across the center: two under the outer edges and two under the deeper rails. The inner pair of end slots attach to the chassis mounts. The rear outer pair of slots, 90 mm apart, belongs to the adjustable bumper.

IF YOU ARE PRINTING THE FLOOR NOW
PRINT FIX: the refreshed floor STLs have their flat mounting faces DOWN on the bed, with reinforcing rails UP. The broad floor web is now 7 mm thick, retaining the same upper mounting datum and underside contacts. Use outer brim only; supports are not needed beneath the main floor. After printing, turn the halves over so insert entrances face up for installation and assembly.

Join the two floor halves

All four part-03 plates bridge the center seam, with one screw in each floor half.

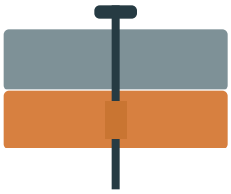


PARTS + HARDWARE
01 + 02 + four insert-equipped 03 plates. Eight M3 x16 screws and head washers.

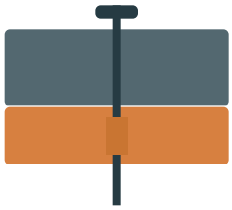
4 **Start all eight screws loosely**
Pass each screw down through the floor into a splice insert. The insert face sits against the floor. Each plate's long direction runs front-to-rear across the seam.

5 **Seat the frame, then tighten progressively**
Hold both halves flat. Snug the four connections a little at a time, alternating sides. The inner plates sit 2 mm lower than the outer plates; v06 adds broad underside seats beneath the outer pair.

OUTER: M3 x16



INNER: M3 x16



Side sections are schematic. The orange plates contain the threads; the bottom rectangle represents the insert. Do not add nuts. Confirm screw-tip clearance on the truck.

Identify the adapter holes

The front and rear adapters use two different hole rows. Do not attach the car through the round row.

16A FRONT

Nominal chassis spacing: 36 mm



SLOTS: screws into the truck

ROUND: inserts for floor screws

Outside end above; vehicle center below.

16B REAR

Nominal chassis spacing: 86 mm



SLOTS: screws into the truck

ROUND: inserts for floor screws

Outside end above; vehicle center below.

6. **Align the slotted row with the truck holes**
   Use 00C/00D templates or dry-fit the actual 16A/16B adapters unloaded. Put the round-hole row inward, toward the center of the chassis.

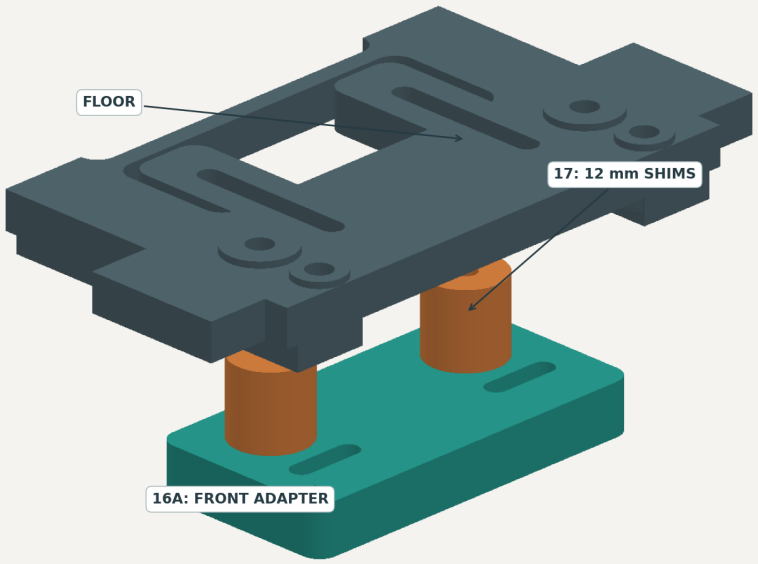
7. **Check support and insert access**
   The plate must bear on sound mounting surfaces. Inspect front recesses and raised lips. If it rocks or rests across uneven bosses, a fitted spacer is needed; tightening harder will bend the plate.

8. **Choose the chassis screws**
   Match the original thread and head style. Allow for the 8 mm adapter thickness while preserving safe thread engagement. The round-row inserts are installed from the top before attaching the floor. Keep chassis slots free of inserts.

WHAT THE SLOTS CAN ADJUST
Photo readings are approximate. Front transverse range: about 27.4-44.6 mm; rear: 77.4-94.6 mm center-to-center. Each floor end can shift about 11.3 mm fore/aft from its nominal position. Slots cannot correct uneven mounting height.

Dry-fit the bare frame

Establish mounting height before adding the upper cart. Keep the bed and hood off for access.



PARTS + HARDWARE
Use your fitted chassis mounts and shims. Where an adapter is used: M3 x25 floor screws into its round-row inserts. Preserve actual front direct-mount screws.

9 **Retain your fitted chassis mounting arrangement**
Keep the front spacer-only arrangement you fitted. At the rear, use the fitted adapter and chassis fasteners. If using an adapter, its round row points inward. The image also shows the optional front adapter.

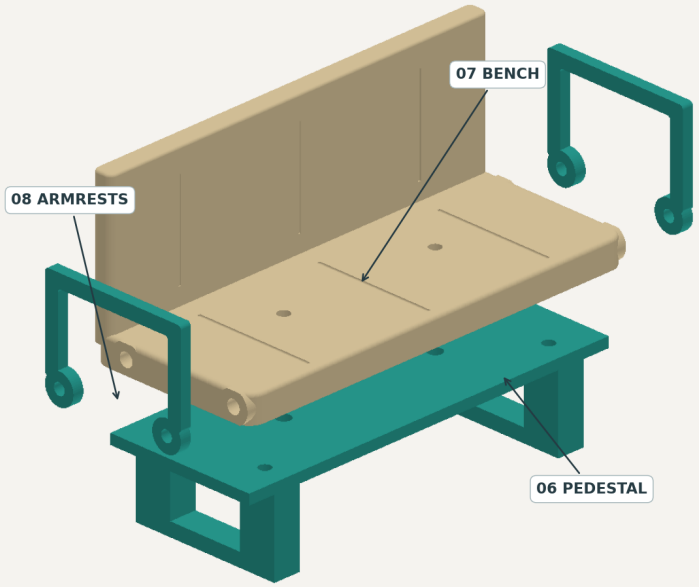
10 **Add shims and lower the floor**
On an adapter, place a shim over each round insert. Pass an M3 x25 down through a floor slot and shim into that insert. At the front, use your fitted direct mounting screws and spacers. Center the floor relative to the wheels.

11 **Check the complete underside, then lift the frame off**
Check motor, wires, inner splice bolt tips and steering at full left/right and compressed suspension. Adjust height if needed. Mark mounting positions; remove the floor mounting screws and return the floor to the bench.

STACK AND HEIGHT
Bottom to top: chassis mounting surface > 8 mm adapter > 12 mm shim > 7 mm floor web. Deep rails extend 4 mm below the underside of the web. This is a trial height, not a verified clearance. Match shim stacks left/right at each end; changing height may require longer frame bolts.

Build the seat assembly

Attach the armrests while you can still reach inside the hollow cushion.



PARTS + HARDWARE
06 pedestal, 07 bench, two 08 armrests. Pedestal: four M3 x30. Armrests: four M3 x8. Bench: two M3 x16. All threads are preinstalled inserts.

12 **Bolt the pedestal to the floor**
Align its four holes in the wide cab area. The pedestal's narrow sides run front-to-rear; its long span runs across the cart. Fit four x30 screws down through the pedestal into the floor inserts.

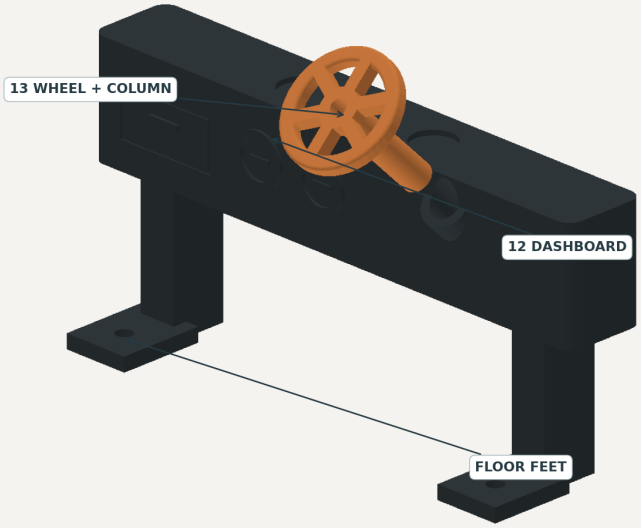
13 **Attach both armrests to the loose bench**
Align the two lower lugs on each armrest with the matching side holes in the cushion. Pass x8 screws inward into the bench side inserts. Use a head washer where it fits flat.

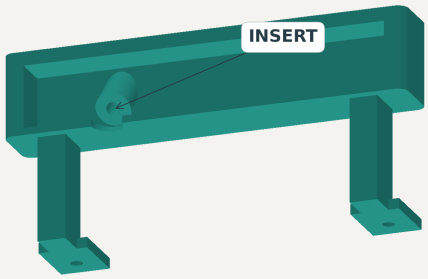
14 **Mount the bench on the pedestal**
The seat back faces the rear bed; the seated position faces the dashboard. Pass two x16 screws through the cushion's internal compression bosses into the pedestal top inserts.

CHECK
The armrests should stand square without pulling the thin cushion walls inward. The bench bolts must bear on the internal bosses, not unsupported hollow walls.

Dashboard and steering wheel

The steering wheel is a fixed decorative part; it does not control the RC steering.





PARTS + HARDWARE
12 dashboard + 13 wheel/column. Feet: two M3 x8 into floor inserts. Column: one M3 x35 into the preinstalled rear boss insert.

15 **Mount the dashboard**
Set the dashboard just ahead of the seat. The gauges and steering boss face the seat; the open rear cavity faces the hood. Bolt both foot tabs to the matching floor holes.

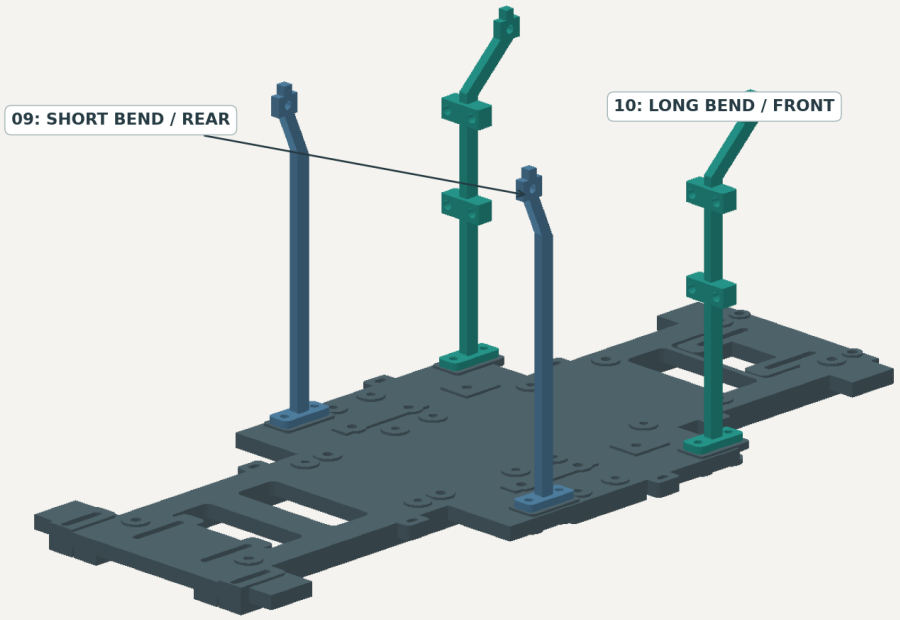
16 **Insert the column**
Clean the axial hole through the wheel hub/column and the angled dashboard socket. Slide the column into the socket without forcing it sideways.

17 **Install the long axial bolt**
Insert the x35 screw through the wheel hub and down the column into the angled boss insert. The insert was installed from the open hood side, before assembly. Tighten gently until seated.

LEAVE ACCESS OPEN
Confirm the steering insert is already installed and cool. Check screw engagement and rear tip clearance before fitting the hood carrier or pumpkin.

Install the four roof posts

Rear posts have the shorter top bend. Front posts have the longer bend and windshield ears.



PARTS + HARDWARE
Two 09 rear posts + two 10 front posts. Eight M3 x8 screws and washers into floor inserts: two per foot.

18 **Position the rear posts**
Place the two short-bend posts behind the seat. Their top ends bend toward the rear bed. Insert each bottom locating peg into its square floor socket.

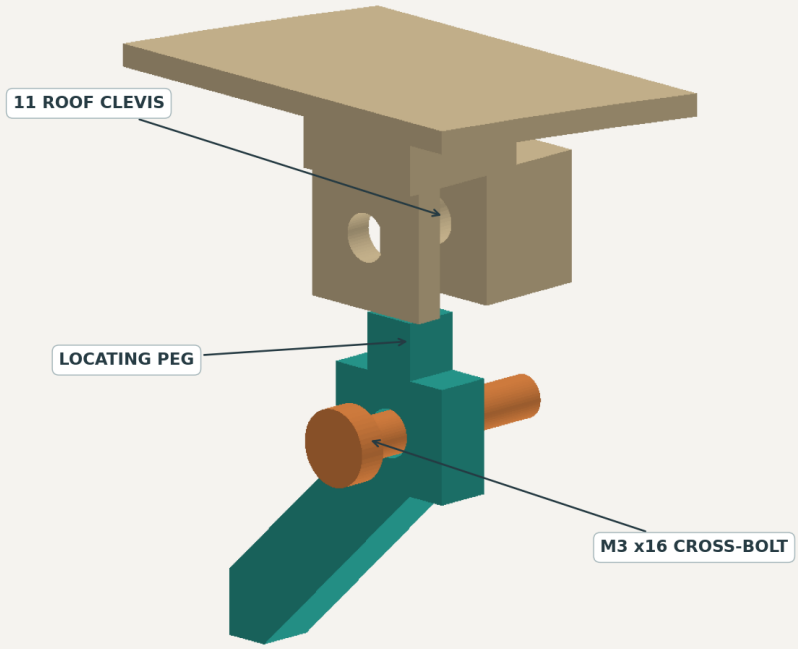
19 **Position the front posts**
Place the two long-bend posts beside the dashboard. Their tops bend forward toward the hood. The windshield ears should line up across the cart.

20 **Bolt the feet, leaving slight adjustment**
Install two x8 screws per foot into the floor inserts. Seat the locating pegs by hand; clean tight sockets instead of forcing them. Leave the bolts slightly loose until the roof aligns all four tops.

CHECK
The posts should stand symmetrically. The top pegs locate the roof; the foot bolts hold the posts down. The reinforced floor can leave the bottom locating pegs recessed below the socket opening.

Fit the roof and cross-bolts

R2 crown and spikes face up. Its four clevis joints use the original mounting interface.



PARTS + HARDWARE
R2 11 roof (or existing roof) + four M3 x16 cross-screws and washers. Threads are in the thicker outside roof cheeks. Illustration shows a corner separated vertically.

21 **Lower the roof evenly**
Align all four top pegs with the square roof sockets. Guide each post tab between its two clevis cheeks. Lower gently until all four corners are seated.

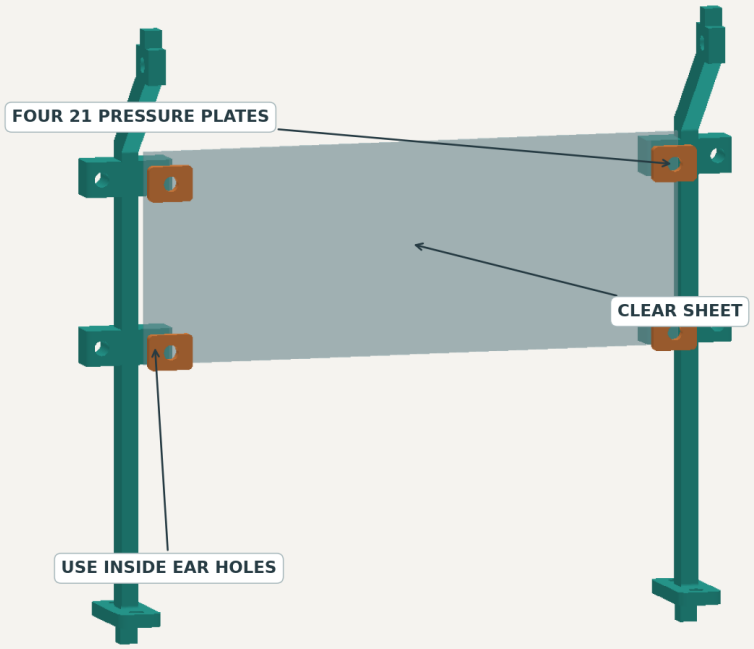
22 **Install the four horizontal bolts**
Start each x16 screw from the thin inward cheek, pass through the post tab, and engage the insert in the thicker outward cheek. The screw head stays toward the cart center.

23 **Square the roof, then snug the joints**
Start all four cross-bolts before tightening. Once the roof sits without twisting, snug the post feet and then the roof bolts. Avoid squeezing the clevis cheeks inward.

REMOVING THE ROOF
Remove the four horizontal cross-bolts and lift the roof vertically off its locating pegs. If a corner will not seat, check support residue in the socket or clevis gap before applying pressure.

Clamp the lower windshield

Use the four inward-facing ear holes. The outer holes are unused.



PARTS + HARDWARE
Four 21 pressure plates, four M3 x10 screws, and clear sheet about 0.5-1.0 mm thick. Starting blank: 124 x48 mm; verify against the assembled posts.

24 **Fit and mark the clear panel**
Hold the sheet against the front faces of the lower front-post ears. Its upper edge should end just below the bends. Mark the four inward holes from the actual assembled posts.

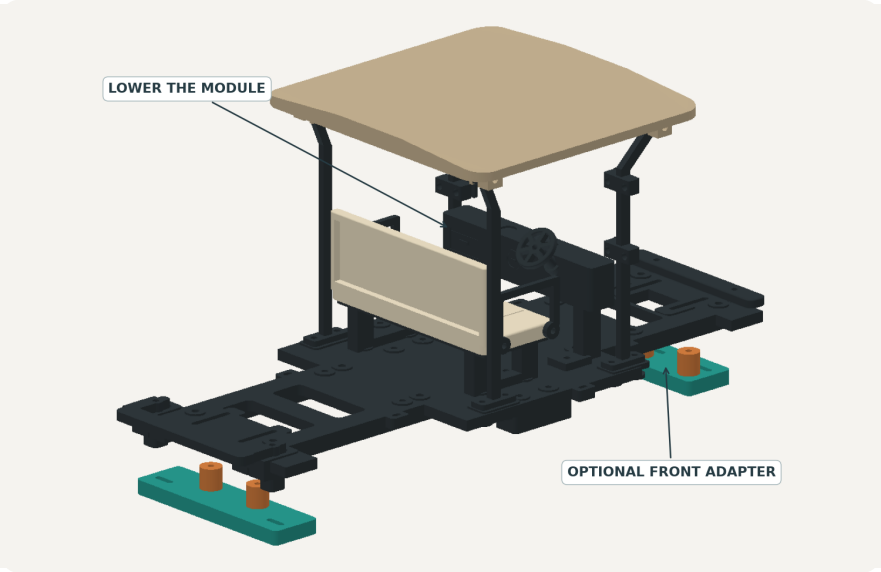
25 **Make clean holes and fit the plates**
Remove the sheet to make the holes, supporting it to avoid cracking. At each location the order is screw head > pressure plate > clear sheet > post ear insert. Inserts enter the ears from the seat side; screws enter from the hood side.

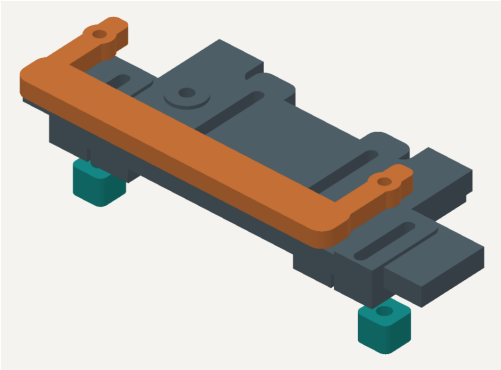
26 **Tighten evenly and lightly**
Start all four bolts before tightening. Use only enough pressure to retain the sheet. Verify screw length and tip clearance with your actual sheet and washer thickness.

ILLUSTRATION
The four orange plates are pulled forward to show where they go. During assembly they sit directly against the clear sheet. The nominal clear panel is a lower windshield, not a full-height panel to the roof.

Mount the cart module

Keep the rear bed and hood carrier off until the four frame-to-adapter bolts are secured.





PARTS + HARDWARE
20A front bumper: two M3 x10. 20B rear bumper + two 20C blocks: two M3 x20. Reuse your fitted chassis mounts; adapters use x25 screws into inserts.

27 **Install the front bumper and rear sliding bumper**
Front: x10 screws through 20A into the floor inserts. Rear: hold 20C blocks insert-face up beneath the outer slots, point 20B arms forward, and start x20 screws through the arms and slots into the blocks. Slide clear of the truck hardware, then snug.

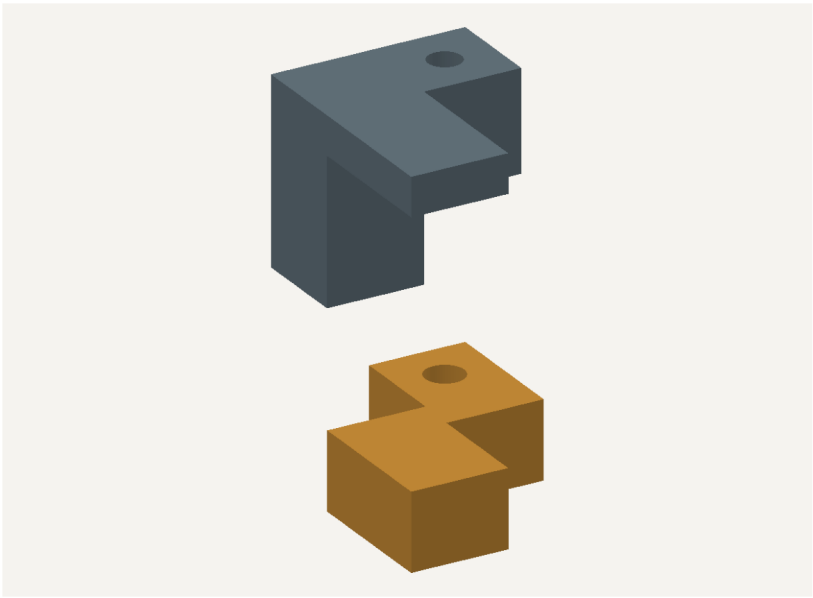
28 **Lower the assembled module onto your fitted mounts**
Keep your fitted spacers in place. At an adapter, refit x25 screws through the inner long floor slots and spacers into the round-row inserts. At the front, retain your direct mount screws. Align and tighten gently.

29 **Repeat the steering and clearance check**
Check both steering extremes and compressed suspension. Look beneath the center splices and both bumpers. Make sure the battery, switch, connectors and mounting screws remain reachable.

REAR BUMPER ADJUSTMENT
Rear slots are 20 x3.4 mm, giving 16.6 mm screw-center travel. Blocks sit beneath the reinforced outer slots, not beneath the inner chassis slots. The new rear bumper replaces the old one. Check the selected position on the actual truck.

Fit the cab clamps first

The clamps grip the outer edge of the flat 7 mm floor web.



PARTS + HARDWARE
Four SR1 03 upper clamps + four SR1 04 lower jaws. Four M3 x16 screws and washers. All eight clamp inserts should already be installed.

30 **Locate the two positions on each side**
From the floor center seam, measure 48 mm toward the rear and 38 mm toward the front. These are clamp/panel-hole centerlines, 86 mm apart. The offset clamp-tightening ear is not the measurement point.

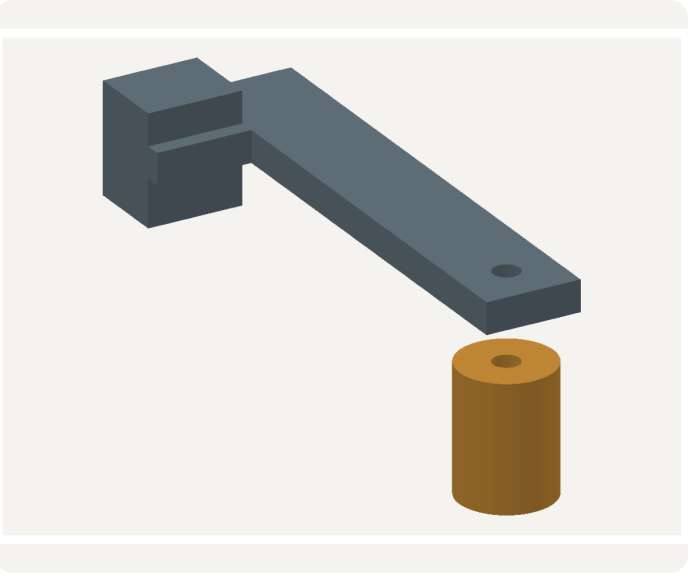
31 **Sandwich the floor edge**
Upper jaw sits above the floor, lower jaw beneath it; the panel insert points outward. Feed one x16 screw down through the upper offset ear into the lower-jaw insert. Snug gently with both jaw faces flat.

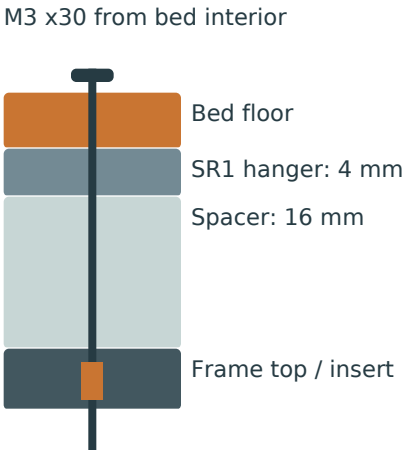
32 **Check clamp grip without forcing it**
Try a light hand movement. It should not slide or rock. If the floor differs from the 7 mm revision or supports prevent seating, correct the fit before loading it. Do not crush the floor to make an incorrect clamp fit.

LEAVE THE PUMPKIN PANELS OFF
Keeping the skirts off gives access to the bed hardware and wires. Both clamps on a side must face the same way and their outward inserts must line up with that skirt.

Install the bed and hangers

This is the updated bed mounting stack for the SR1 rear arches.





PARTS + HARDWARE
R2 04 bed; four SR1 07 spacers (16 mm); two SR1 05 rear hangers; two SR1 06 forward hangers. Four M3 x30 + washers into rear-floor inserts.

33 **Place spacers and the correct hanger pairs**
Rear floor holes are X=-150 and -48, Y=+/-30. Put 16 mm spacers on all four. SR1 05 hangers go at the rear pair. SR1 06 go at the forward pair; their offset ends point rearward. All insert bosses point outwards.

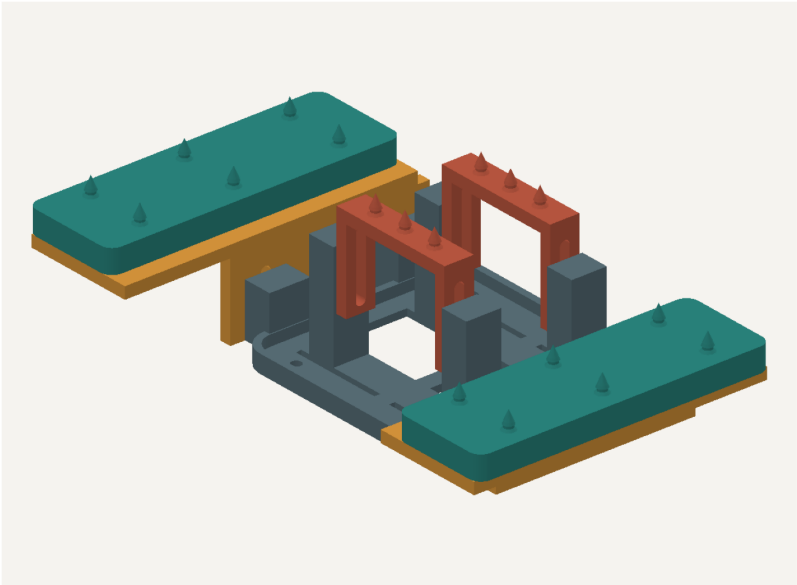
34 **Set the bed over the hangers**
Bottom to top: floor, 16 mm spacer, 4 mm hanger plate, bed. The bed wall without spikes faces the seat. Start all four x30 screws through the bed and hangers into the floor inserts, then snug evenly.

35 **Confirm height and service access**
The 16+4 mm support stack replaces the earlier 20 mm spacer. If your actual spacers differ, preserve their original height by subtracting 4 mm for each hanger. Verify all four spacers sit flat and bolts clear the chassis.

DO NOT STACK THE HANGERS ON THE OLD 20 mm SPACERS
That would lift the bed by 4 mm and change bolt engagement. No extra nuts are used; the threads remain in the floor.

Mount the raised hood carrier

Two adjustable bars reach the flesh inside the hollow pumpkin hood.



PARTS + HARDWARE
HF2 01 carrier, two HF2 05 spike bars, four original 9 mm spacers. Base: four M3 x16. Bars: four M3 x10 + washers.

36 **Bolt the bare carrier to the front floor**
Position the 9 mm spacers at X=100 and 160, Y=+/-30. Place the HF2 carrier above them and fit four x16 screws through its 60 x60 mm mounting pattern. Keep the tall towers upright.

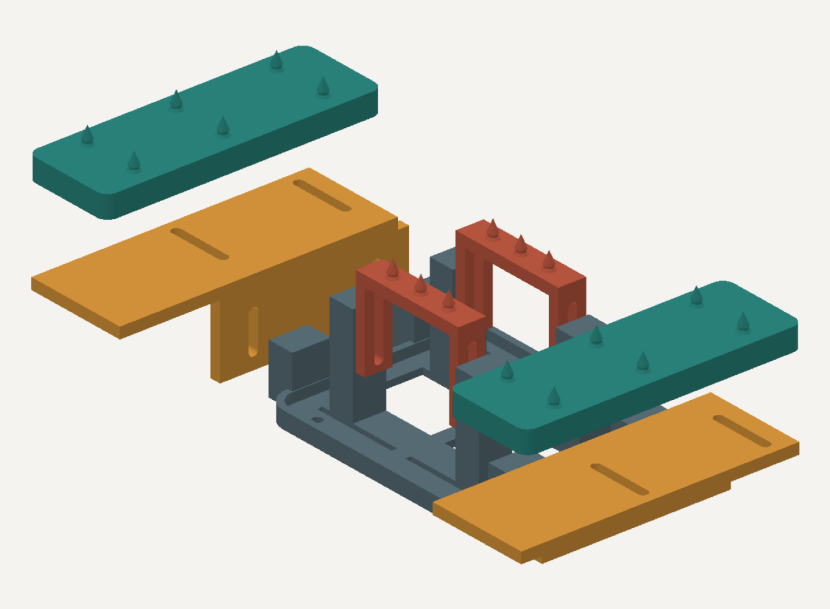
37 **Fit both U-shaped spike bars**
Each bar spans across the cart between a pair of towers, with its spikes up. Insert x10 screws from INSIDE the U through its two vertical slots into the inward-facing tower inserts. Use a washer under each head.

38 **Set both bars near the middle of travel**
Start with the contact surfaces about 50 mm above the carrier bottom. Leave adjustment possible, but keep screws engaged. Final range is 40-60 mm; spike tips are 6 mm above the contact surfaces. Fixed towers end at 38 mm.

HOLLOW HOOD FIT
The pumpkin will be supported at its inner ceiling, not at the empty cavity bottom. Remove the pumpkin to adjust and tighten the bars. If the inner ceiling is above the 60 mm seats, taller supports are needed; do not leave the shell floating above the spikes.

Attach the two front fender pads

The long end of each pad and shelf points toward the seat.



PARTS + HARDWARE
HF2 02/03 mirrored risers + two HF2 04 pads. Four M3 x10 + washers for the risers; four M3 x10 + washers for the pads.

39 **Bolt the risers to the carrier sides**
Place a riser against each outside face. Its shelf projects outward. Pass two x10 screws through the vertical slots into the low outside carrier inserts. Start at mid-height.

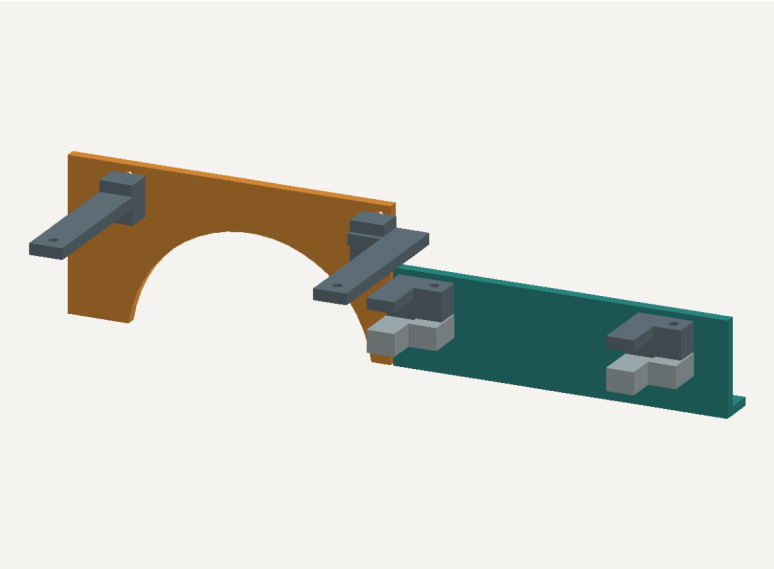
40 **Bolt each pad down from underneath**
Place a pad on each shelf, spikes up and longer end rearward. From BELOW the shelf, fit two x10 screws with washers through the transverse slots into the underside pad inserts.

41 **Adjust width and height unloaded**
Each riser has 12 mm total height travel. Each pad has 20 mm lateral travel; its bolt centerline can sit 65-85 mm from the center of the cart. Tighten both fasteners in each adjustment after checking tire clearance.

ONE SHELL OR THREE PIECES
The center bars and two pads can carry either one connected pumpkin shell or separate hood/left-fender/right-fender pieces. Do not assume their seats lie in one plane; carve and adjust to suit the actual underside.

Hang the skirts and rear arches

The printed backing supports the pumpkin; the slots attach to the brackets.



PARTS + HARDWARE
Two SR1 01 cab skirts and two SR1 02 rear arches. Eight M3 x10 + washers at zero outward offset. Optional 5/10 mm stand-offs require the screw stacks listed earlier.

42 **Attach the cab skirts**
Spikes and ledges face away from the cart. From OUTSIDE, put two x10 screws through each skirt vertical slots into the cab-clamp inserts. Hole spacing is 86 mm. Start at mid-height.

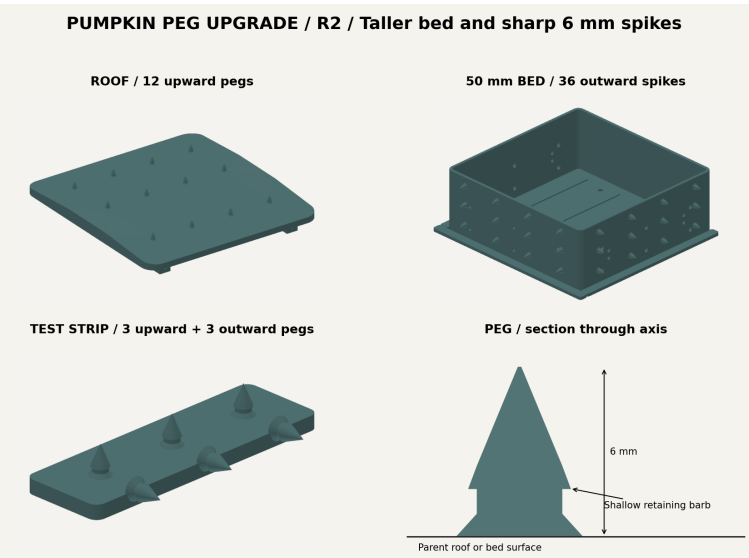
43 **Attach the rear arch panels**
Align the wheel opening over the rear tire. From outside, put two x10 screws through the panel slots into the bed-hanger inserts. Hanger centers are X=-150 and -60: 90 mm apart.

44 **Set height and outward clearance**
Each panel has 10 mm total vertical travel. If more outward space is needed, place equal 5 or 10 mm stand-offs behind BOTH slots of that panel and use the matching longer screws. Snug all screws with washers bridging the slots.

CHECK THE REAL TIRE ENVELOPE
The rear arch is R44 around an assumed wheel center; it is not a measured full-travel envelope. Turn steering fully and compress each corner. A printed backing that rubs must be readjusted or revised before adding pumpkin.

Lights, plaques and moisture protection

Do this while every support and fastener is still visible.



45 **Clean and inspect every spike**
Remove support residue gently while supporting each spike root. Check the bed side/rear spikes, roof spikes, hood bars, fender pads and runner panels. A broken or soft spike may need repair/reprinting before carving.

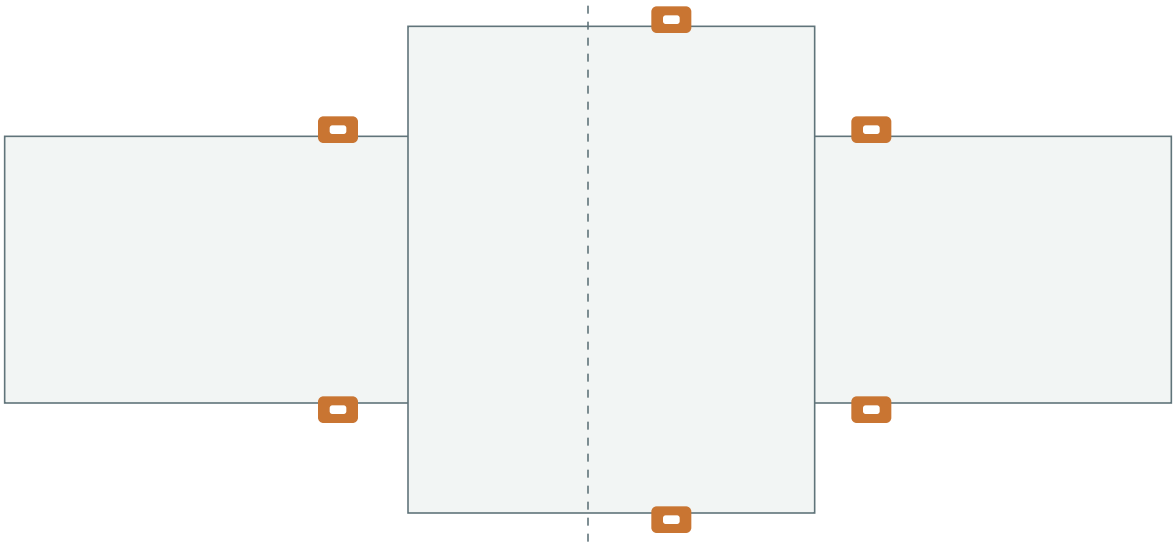
46 **Prepare plaques and light bezels**
Keep the two IT DEPT plaques and two LED bezels ready for the pumpkin. Test-fit LEDs and route their leads now; their small ear holes can take soft thread for attachment. Do not guess LED voltage or connect an unknown LED directly to the RC battery.

47 **Add a removable moisture barrier**
Protect electronics from drips while keeping motor cooling, connectors and the battery accessible. The printed bed is not watertight. Keep mounting screw access open and avoid trapping wires under the bed or hood.

> **STOP BEFORE ATTACHING PUMPKIN**
> Finish wiring, optional cargo and the dry-assembly checks on the following pages first. Add fresh pumpkin later, using the final readiness page as the handoff.

Secure wiring to the eyelets

Six integral tie points keep loose bundles near the frame and away from moving parts.



Orange markers are the six 5 x2.6 mm tie eyelets. This top view is schematic; use the actual truck to select a route clear of the steering, motor, tires and suspension.



TIE PATH / SCHEMATIC
Feed through the eyelet, loop around the bundle and outside the eyelet, then latch. Orange = tie. Teal = wire bundle.

48 **Route while the underside remains visible**
Keep connectors and the switch reachable. Leave slack at plugs and at the removable hood lights. Avoid resting insulation against motor/drivetrain parts or sharp screw ends.

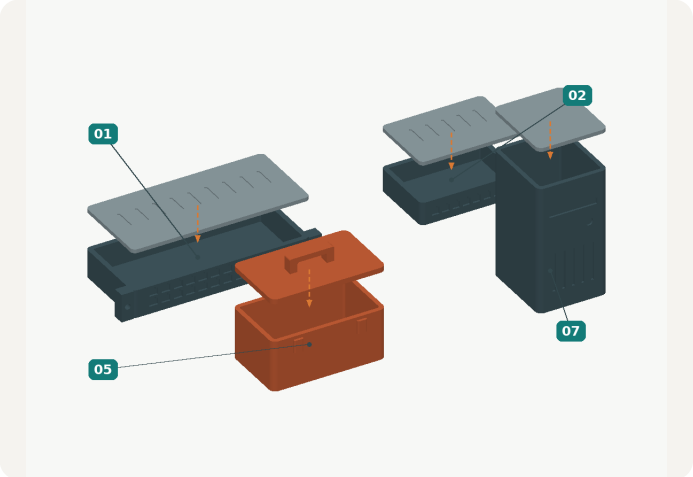
49 **Snug the ties without crushing insulation**
Use narrow ties that pass freely through the slots. Secure the bundle without stretching it. Trim tails flush and turn latch heads away from nearby tires and moving links.

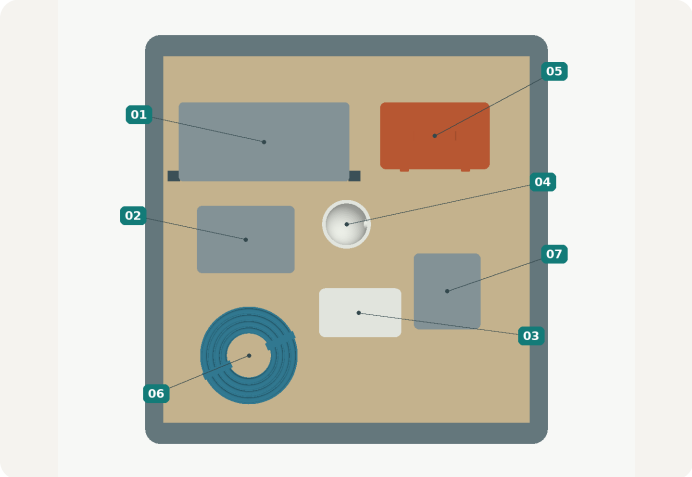
50 **Check the full movement range**
With power off, turn fully left/right and gently compress the suspension at each end and opposite corners. Confirm wires, ties and latch heads stay clear. Recheck after adding the pumpkin and cargo.

EYELETS ARE FOR WIRING
Do not use them to lift the cart or retain the pumpkin/cargo. Use the bed tie holes for cargo. Keep fresh pumpkin moisture away from electrical parts.

Assemble and secure the IT cargo

Optional Cargo_v02 parts fit the existing bed. Use as many as the tested load allows.





51 **Match the bodies and lids**
The 24-port switch, 8-port switch, toolbox and optional PC tower each have separate lids. The bullet camera, dome camera and cable coil are single-piece props.

52 **Retain the flat lids**
Dry-fit and center each lid on its rim. These are flat lids, not snap-fits. Use two narrow removable bands over each lid and under its body. On the toolbox, place bands on either side of the handle.

53 **Build a removable cargo insert**
Cut a light insert about 120 x120 mm, then trim to fit the real bed. Make paired holes beside the props. Tie each prop to the insert; a case band can retain both its lid and the whole case.

54 **Tie the insert to the bed**
Use accessible bed tie holes. Keep the mounting screws serviceable and the moisture barrier effective. Check that no prop or lid slides free when the cart is gently tilted.

OPTIONAL FINISH
Paint ports and lenses for contrast. The toolbox handle is decorative; do not lift the model by it. Include the insert, ties, paint and all cargo in the final weight check.

Inspect the completed dry build

Component CAD checks are complete. Final combined fit on the real truck still needs this check.

Check	What to verify
Frame and joints	Both floor halves sit flat. All four splice plates are installed. No cracked PLA, loose screws, spinning inserts or crushed pads.
Mounts and shims	Your fitted mount stack is seated on sound surfaces. Keep the direct front mount. Chassis screws have correct thread and safe engagement.
Steering and suspension	Turn fully left/right and compress each end and opposite corners. Tires, links and shafts clear the floor, screw tips, ties and wires.
Electrical and battery access	Battery is secure and removable. Wires cannot rub the motor/drivetrain or be pinched by the floor. Switch remains accessible.
Upper assembly	Seat, roof, windshield and bumpers are secure. All empty pumpkin supports are secure. Cargo and lights cannot fall free.

55

Weigh and test progressively

Test the dry cart at low speed first. After pumpkin is attached, record the full display weight and test again gradually. Watch steering effort, suspension and motor behavior; no load rating has been established.

Symptom	Next action
Adapter rocks or bolts miss	Recheck orientation, slot alignment and boss height. Send a close-up; do not pull it flat with the screws.
Roof corner will not seat	Loosen post feet slightly, remove socket/clevis debris, then align all four corners together.
Frame flexes, tires rub or drive struggles	Stop and reduce the load. Check fasteners, ride height and clearances before continuing.
Windshield cracks or bows	Back off clamp pressure; check hole alignment and support around each fastener.

AFTER THE FIRST SHORT RUN

Recheck the frame and mounting screws, look for fresh rub marks, and verify that the pumpkin and cargo have not shifted. The stronger frame does not increase the truck's drivetrain or suspension capacity.

READY FOR PUMPKIN

Final dry-build handoff

Everything below should be checked before you press fresh pumpkin onto the spikes.

Tick	Verify
[ ]	Frame: four splices installed; fitted chassis mounts secure and level; original front direct-mount arrangement retained.
[ ]	Seat, arms, dashboard, column, posts, roof and windshield secure; no spinning inserts, cracks or stressed clevises.
[ ]	Bed: hanger plates fitted over the correct spacers; forward hangers offset toward rear; bed height preserved.
[ ]	HF2: two sliding bars and two fender pads secure; slots retain adjustment; all four base spacers flat.
[ ]	SR1: four clamps grip without crushing; cab skirts and rear arches secure; spike faces point outward.
[ ]	Full steering and suspension movement clear the brackets, panels, bolts and wiring. Battery and switch remain reachable.
[ ]	Loose wires secured; lights tested with their correct supply; barrier protects electronics without obstructing cooling.
[ ]	Optional cargo restrained; all remaining pumpkin-contact spikes intact; screws accessible for later removal.

PUMPKIN ATTACHMENT ORDER

1. Bed side pieces, then rear piece on the ledges.
2. Cab skirts and rear arches, seated on their ledges.
3. Hood and fenders: trial-fit, adjust supports to the inner flesh, remove shell to tighten, then press on.
4. Roof rind last, supported against the crown.
5. Plaques and light bezels; reconnect removable light plugs and repeat clearance checks.

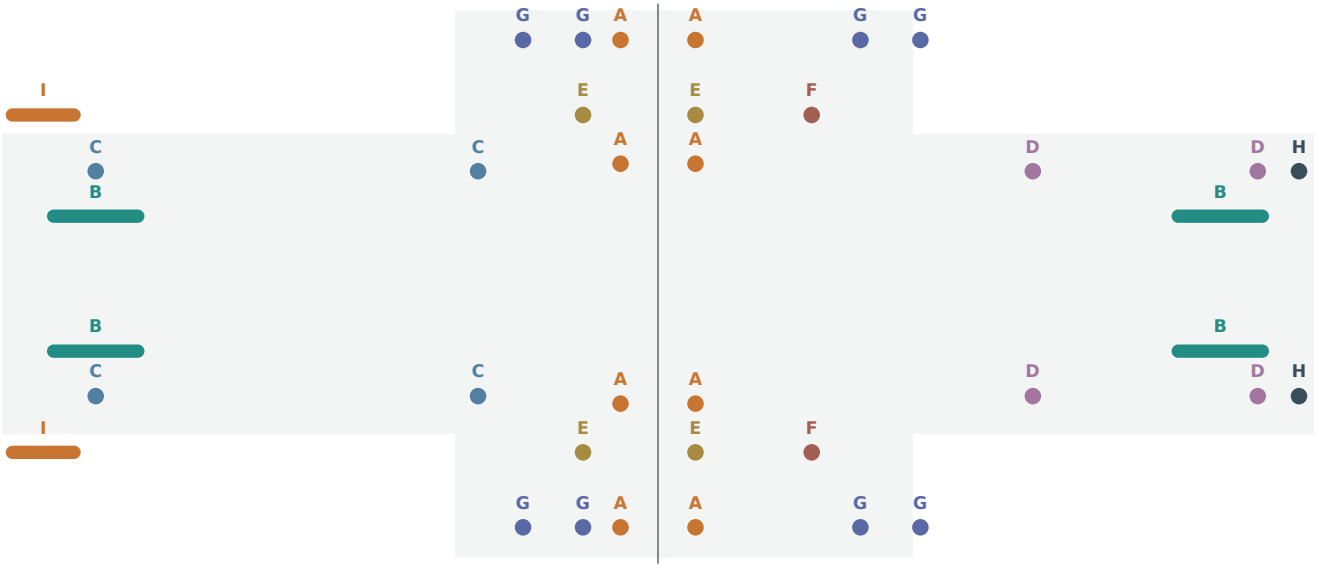
Start with about **8-10 mm or more of flesh at the 6 mm spike locations**. Test a scrap before committing carved pieces. Leave room for the roof curve and the hollow hood interior supports. Do not use the obsolete 30 mm-high hood template. Grip changes as pumpkin softens, so check it again before the event.

At this point the dry cart is complete and ready for pumpkin. After pumpkin is attached, repeat the movement check and make a short, low-speed test with the final load.

REFERENCE

Floor hole map

All coordinates are millimeters in the CAD model. Rear is -X; front is +X. Use this map to distinguish nearby holes.



ID	Connection	X positions	Y positions
A	Four splice plates	-10, +10	-65, -32, +32, +65
B	Adapter slots (nominal centers)	-150, +150	-18, +18
C	Bed spacers	-150, -48	-30, +30
D	Hood spacers	+100, +160	-30, +30
E	Seat pedestal	-20, +10	-45, +45
F	Dashboard feet	+41	-45, +45
G	Roof-post foot bolts	-36, -20, +54, +70	-65, +65
H	Front bumper / floor inserts	+171	-30, +30
I	Rear bumper / insert blocks	-164 slot centers	-45, +45

Square post sockets are at X = -28 and +62, Y = +/-65. The old central chassis holes at X = +/-60, Y = +/-30 remain present but are unused with the new end adapters. Only C/D/E/F/G/H receive floor inserts. A uses splice inserts, B adapter inserts, and I slider-block inserts. This is not a drilling template.

Guide basis: v06 flat-floor source; R2 peg upgrade; HF2 hood/fender CAD; SR1 runner CAD; Cargo_v02; user chassis photos. Core diagrams show unchanged connections; current upgrade diagrams show their separate assemblies. Full combined physical fit remains unverified.